Lab 3

MULTI QUEUE Scheduling

```
#include<stdio.h>
#define MAX 20
struct Process{int pid,at,bt,ct,tat,wt,type;};
int main(){
struct Process p[MAX],sq[MAX],uq[MAX];
int n,i,sq_count=0,uq_count=0,time=0;
printf("Enter number of processes: ");
scanf("%d",&n);
for(i=0;i<n;i++){
printf("\nProcess %d\n",i+1);
p[i].pid=i+1;
printf("Arrival Time: ");
scanf("%d",&p[i].at);
printf("Burst Time: ");
scanf("%d",&p[i].bt);
printf("Type (1=System,2=User): ");
scanf("%d",&p[i].type);
if(p[i].type==1) sq[sq_count++]=p[i];
else uq[uq_count++]=p[i];
}
for(i=0;i<sq_count;i++){
if(time<sq[i].at) time=sq[i].at;
time+=sq[i].bt;
sq[i].ct=time;
}
for(i=0;i<uq_count;i++){
if(time<uq[i].at) time=uq[i].at;
time+=uq[i].bt;
uq[i].ct=time;
}
float total_tat=0,total_wt=0;
printf("\n---System Queue---\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for(i=0;i<sq_count;i++){
sq[i].tat=sq[i].ct-sq[i].at;
sq[i].wt=sq[i].tat-sq[i].bt;
total_tat+=sq[i].tat;
total_wt+=sq[i].wt;
```

```
printf("%d\t%d\t%d\t%d\t%d\t%d\n",sq[i].pid,sq[i].at,sq[i].bt,sq[i].ct,sq
[i].tat,sq[i].wt);
}
printf("\n---User Queue---\n");
printf("PID\tAT\tBT\tCT\tTAT\tWT\n");
for(i=0;i<uq_count;i++){
uq[i].tat=uq[i].ct-uq[i].at;
uq[i].wt=uq[i].tat-uq[i].bt;
total_tat+=uq[i].tat;
total_wt+=uq[i].wt;
printf("%d\t%d\t%d\t%d\t%d\t%d\n",uq[i].pid,uq[i].at,uq[i].bt,uq[i].ct,u
q[i].tat,uq[i].wt);
}
printf("\nAverage Turnaround Time=%.2f",total_tat/n);
printf("\nAverage Waiting Time=%.2f\n",total_wt/n);
printf("Nidhi R Khabiya\n");
return 0;
}
```

Output

```
Output                                                    Clear
Enter number of processes: 4

Process 1
Arrival Time: 0
Burst Time: 4
Type (1=System,2=User): 1

Process 2
Arrival Time: 0
Burst Time: 3
Type (1=System,2=User): 1

Process 3
Arrival Time: 0
Burst Time: 8
Type (1=System,2=User): 2

Process 4
Arrival Time: 10
Burst Time: 5
Type (1=System,2=User): 1

---System Queue---
PID AT  BT  CT  TAT WT
1   0   4   4   4   0
2   0   3   7   7   4
4   10  5   15  5   0

---User Queue---
PID AT  BT  CT  TAT WT
3   0   8   23  23  15
```

```
---System Queue---
PID AT  BT  CT  TAT WT
1   0   4   4   4   0
2   0   3   7   7   4
4   10  5   15  5   0

---User Queue---
PID AT  BT  CT  TAT WT
3   0   8   23  23  15

Average Turnaround Time=9.75
Average Waiting Time=4.75
Nidhi R Khabiya


=== Code Execution Successful ===
```